

Yue Xue (薛岳)

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<https://scholar.google.com/citations?user=6cdLQQQAAAAJ>

SUMMARY

- Senior Security Engineer focused on blockchain security, software security, and LLM applications in security.
- An early LLM4Sec researcher and builder: co-author of foundational, highly-cited works (GPTScan, PropertyGPT), advancing a security methodology combining LLMs with program analysis/formal verification, and building Finite Monkey to industrialize AI smart contract auditing at scale.
- Owned internal anti-money laundering (AML) initiatives at OKX with end-to-end on-chain compliance experience, from fund monitoring and risk identification to alerting/interdiction and closed-loop handling.
- Experienced in smart contract auditing, static analysis, formal verification, and on/off-chain monitoring systems.
- Awards: ETHSecurity Badge holder; NDSS 2025 Distinguished Paper Award; ACM SIGSOFT Distinguished Paper Award (FSE 2024); Secure3 Rank #2 (total bounty USD 68,570).

EDUCATION

Guangzhou University (GZHU) <i>M.S. in Computer Technology (Academician Fang Binxing Class)</i> · Research: blockchain security, smart contract security, LLM for security	2020.09 – 2023.06
University of Electronic Science and Technology of China (UESTC) <i>B.E. in Electrical Engineering and Automation</i> · Focus: electrical engineering, automatic control, and core engineering coursework	2011.09 – 2015.06

EXPERIENCE

CertiK <i>Head of AI Research</i> · Leading AI vulnerability scanning engine development	2025.11 – Present
OKX <i>Senior Security Engineer</i> · Led a 4-person team to build internal contract security monitoring (state and transaction anomaly detection) · Designed and maintained the internal contract management platform and privileged-task framework; delivered safe privilege migration with zero incidents · Built AML monitoring for CeFi deposits and wallet transactions; continuously improved black-fund detection and interception · Conducted smart contract auditing and security tooling development	2022.04 – 2022.10, 2024.05 – 2025.11
MetaTrust <i>Security Supervisor / Consultant</i> · Led smart contract audits and engagement coordination; improved audit workflows · Developed 60+ Slither detectors and optimized nearly 100 rules; reduced false positives below 25% · Contributed to GPTScan and independently built PropertyGPT for formal verification · Collaborated with NTU research teams and co-authored papers	2022.10 – 2025.08
C# Developer · Built full-stack web features and maintained databases · Developed and maintained enterprise systems	2016.12 – 2019.01

PROJECTS & TOOLS

LLM4Sec Audit Tools <i>End-to-end audit toolchain</i> · Code Audit Hintor (IDE security plugin, 37 stars, 2025.03–Present): RAG-powered audit hints for VSCode and Cursor · Finite Monkey Engine (349 stars, 2024.04–Present): task-driven LLM audit engine with practical precision/recall · TxAnalyzer (GitHub, 2026.05–Present): agent skill and CLI for blockchain attack transaction root-cause analysis, with EVM/Solana artifact collection · AI Audit Result Evaluator (private, 2025.05–Present): automated judging for contest findings vs. ground truth · Code Semantic Search (private, 2024.12–Present): LLM-based codebase understanding and retrieval	Ongoing
Falcon <i>Enhanced Solidity static analysis framework</i> · Extended Slither with dozens of new detectors and improved precision · Supports DeFi price manipulation, centralization risk, and complex logic bug detection	2022.12 – 2024.05

PUBLICATIONS

1. Y. Sun, D. Wu, **Yue Xue**, H. Liu, H. Wang, Z. Xu, X. Xie, Y. Liu. “GPTScan: Detecting Logic Vulnerabilities in Smart Contracts by Combining GPT with Program Analysis.” Proc. ICSE, 2024.
2. Y. Sun, D. Wu, **Yue Xue**, H. Liu, W. Ma, L. Zhang, Y. Liu, Y. Li. “LLM4Vuln: A Unified Evaluation Framework for Decoupling and Enhancing LLMs’ Vulnerability Reasoning.” arXiv preprint arXiv:2401.16185, 2024.
3. W. Ma, D. Wu, Y. Sun, T. Wang, S. Liu, J. Zhang, **Yue Xue**, Y. Liu. “Combining Fine-Tuning and LLM-based Agents for Intuitive Smart Contract Auditing with Justifications.” Proc. ICSE, 2025.
4. Y. Liu, **Yue Xue**^{*}, D. Wu, Y. Sun, Y. Li, M. Shi, Y. Liu. “PropertyGPT: LLM-driven Formal Verification of Smart Contracts through Retrieval-Augmented Property Generation.” NDSS, 2025.
5. K. Li, **Yue Xue**, S. Chen, H. Liu, K. Sun, M. Hu, H. Wang, Y. Liu, Y. Chen. “Static Application Security Testing (SAST) Tools for Smart Contracts: How Far Are We?” Proc. ACM Softw. Eng. (FSE), 1, 1447–1470, 2024.
6. **Yue Xue**, J. Fu, S. Su, Z. A. Bhuiyan, J. Qiu, H. Lu, N. Hu, Z. Tian. “Preventing Price Manipulation Attack by Front-Running.” Proc. AAIS, 309–322, 2022.
7. M. Xie, M. Hu, Z. Kong, C. Zhang, Y. Feng, H. Wang, **Yue Xue**, H. Zhang, Y. Liu, et al. “DeFort: Automatic Detection and Analysis of Price Manipulation Attacks in DeFi Applications.” Proc. ISSTA, 2024.
8. **Yue Xue**, D. Fan, S. Su, J. Fu, N. Hu, W. Liu, Z. Tian. “A Review on the Security of the Ethereum-Based DeFi Ecosystem.” CMES–Computer Modeling in Engineering & Sciences, 139(1), 2024.
9. S. Su, Y. Tan, **Yue Xue**, C. Wang, H. Lu, Z. Tian, C. Shan, X. Du. “Detecting Smart Contract Project Anomalies in Metaverse.” Proc. IEEE MetaCom, 2023.
10. Z. Kong, C. Zhang, M. Xie, M. Hu, **Yue Xue**, Y. Liu, H. Wang, Y. Liu. “Smart Contract Fuzzing Towards Profitable Vulnerabilities.” Proc. ACM Softw. Eng. (FSE), 2, 153–175, 2025.
11. S. Su, **Yue Xue**, L. Lin, C. Wang, H. Lu, J. Qiu, Y. Sun, Y. Liu, Z. Tian. “Smart Contract Firewall: Protecting the On-Chain Smart Contract Projects.” Proc. IEEE GLOBECOM, 5122–5128, 2023.
12. Y. Sun, S. Ding, L. Tan, **Yue Xue**, B. Li, C. Huang, S. Xiong. “Beyond Stars and Commits: A Fair and Manipulation-Resistant Framework for Measuring Developer Contributions.” Proc. ACM FSE Companion, 2025.
13. **Yue Xue**, et al. “Understanding Codebase like a Professional! Human–AI Collaboration for Code Comprehension.” Proc. ICPC, 2026 (accepted).

* Co-first author.

ACADEMIC SERVICE

- Reviewer: IEEE Transactions on Software Engineering (TSE); ACM Transactions on Software Engineering and Methodology (TOSEM); ACM Transactions on Privacy and Security (TOPS)

INDUSTRY WRITING

- Authored widely circulated security and AI research articles, including X posts with tens of thousands of views: [article 1](#); [article 2](#).

AWARDS

- NDSS 2025 Distinguished Paper Award: PropertyGPT
- ACM SIGSOFT Distinguished Paper Award (FSE 2024): SAST Tools for Smart Contracts
- GLOBECOM 2023 Best Paper Award: Smart Contract Firewall
- ETHSecurity Badge holder
- Secure3 Rank #2 (total bounty USD 68,570)